Monte-Carlo statistical uncertainty in the SolarOscFitter likelihood

The Dembinski–Abdelmotteleb and Conway (Barlow–Beeston) treatments, and how they are implemented

Notes for the SolarOscFitter analysis

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Abstract

The predicted event counts in the fit are built from finite Monte-Carlo (MC) samples: signal response matrices and background templates. Their statistical fluctuations are not in a plain Poisson likelihood, which treats the prediction as exact. This note explains the method of Dembinski and Abdelmotteleb [1], which derives a fast approximate likelihood from the exact one of Barlow and Beeston [2], together with Conway's earlier approximation [3]. It then describes how the fitter implements both (`mc_stat_likelihood`), including the adaptation needed because the day bin and all night bins of one reconstructed-energy bin are built from the same MC events. **The paper:** H. Dembinski and A. Abdelmotteleb, *A fast and stable approximate maximum-likelihood method for template fits*, arXiv:2206.12346.

1 The problem

In each analysis bin the fit compares an observed count n with a prediction $\mu(\boldsymbol{p})$ that depends on the parameters $\boldsymbol{p}$ (oscillation parameters, fluxes, nuisances). The plain Poisson likelihood assumes μ is known exactly. But μ is computed from simulation: a response matrix filled with a finite number of MC events, background templates filled with weighted MC events. If the MC sample were generated again, μ would change. When that MC fluctuation is not small compared with the data fluctuation $\sqrt{n}$, ignoring it makes the fit too confident: contours shrink and pulls are too wide. With Asimov data the best fit does not move, but the sensitivity is overstated.

The standard remedy treats the true template content in each bin as unknown, constrained by the MC that was actually generated. The question is how to do that without introducing a free parameter for every template in every bin.

2 Baker–Cousins form

All likelihoods below are written, following Baker and Cousins [4], as

$$Q(\boldsymbol{p}) = -2 \ln \frac{L(n; \mu(\boldsymbol{p}))}{L(n; n)}, \tag{1}$$

whose minimum is asymptotically χ^2-distributed. For one Poisson bin,

$$Q_{\mathrm{P}}(n; \mu) = 2 \left[\mu - n + n \ln \frac{n}{\mu} \right], \tag{2}$$

with the convention $n \ln(n/\mu) = 0$ for $n = 0$. This is exactly what the fitter's `poissonLogLikelihood` computes (as $-\frac{1}{2} Q_{\mathrm{P}}$).

3 The exact Barlow–Beeston likelihood

Let the prediction in a bin be a sum of components (templates) k with yields y_k. Template k has an observed MC count a_k in the bin, out of M_k in total, and an unknown true expected MC count ξ_k. Then $\mu = \sum_k y_k \xi_k / M_k$, and the MC counts are themselves Poisson around ξ_k. The exact likelihood of data and simulation together is [2, 1]

$$Q = Q_{\mathrm{P}}(n; \mu) + \sum_k Q_{\mathrm{P}}(a_k; \xi_k). \tag{3}$$

Every ξ_k in every bin is a nuisance parameter. Barlow and Beeston profile them by solving a non-linear equation per bin. This is exact, but slow when there are many bins, and the finite accuracy of the solver can make the likelihood slightly discontinuous, which upsets gradient-based minimisers and samplers.

4 Conway's approximation ("Barlow–Beeston lite")

Write $\xi_k = \beta_k a_k$. Conway's central approximation [3] is to use one common factor $\beta_k \approx \beta$ per bin:

$$\mu = \sum_k \frac{y_k \beta_k a_k}{M_k} \approx \beta \underbrace{\sum_k \frac{y_k a_k}{M_k}}_{\mu_0} = \beta \mu_0. \tag{4}$$

This holds when one component dominates the bin. Inserting it in Eq. (3) gives ([1], Eq. 4)

$$Q \approx Q_{\mathrm{P}}(n; \beta\mu_0) + 2a\left[(\beta - 1) - \ln\beta\right], \qquad a = \sum_k a_k. \tag{5}$$

Conway then expands $\ln\beta$ to second order around $\beta = 1$, so the constraint becomes Gaussian:

$$Q = Q_{\mathrm{P}}(n; \beta\mu_0) + \frac{(\beta-1)^2}{V(\beta)}, \qquad V(\beta) \approx \frac{1}{a} = \frac{\sum w^2}{\mu_0^2}. \tag{6}$$

$V(\beta)$ is the squared relative MC uncertainty of the bin; with weighted MC it is $\sum_i w_i^2 / (\sum_i w_i)^2$. Setting $\partial Q / \partial \beta = 0$ gives a quadratic with one valid root, so β is profiled in closed form. This is HistFactory's `staterror`, and MaCh3's `kBarlowBeeston`.

The Gaussian expansion is good when a is large, but frequently the MC sample is *not* large in every bin: then the constraint is too weak in the direction of small β and too strong for large β.

5 The Dembinski–Abdelmotteleb approximation

Dembinski and Abdelmotteleb [1] start from the same Eq. (5), but do *not* expand the logarithm. Minimising over β directly,

$$\frac{\partial Q}{\partial \beta} = 0 \;\Rightarrow\; \mu_0 - \frac{n}{\beta} + a\Big(1 - \frac{1}{\beta}\Big) = 0 \;\Rightarrow\; \boxed{\beta = \frac{n + a}{\mu_0 + a}}, \tag{7}$$

which is even simpler than Conway's root ([1], Eq. 11). The interpretation is direct:

- if $a \ll n$ (little MC), β simply moves μ_0 to n, and the bin says nothing about the parameters;
- if $a \to \infty$ (perfect MC), $\beta \to 1$ and $Q \to Q_{\mathrm{P}}(n; \mu_0)$, the plain Poisson likelihood.

With a few lines of algebra, Eq. (5) can be rewritten as ([1], Eq. 12)

$$\boxed{Q = Q_{\mathrm{P}}(n; \beta\mu_0) + Q_{\mathrm{P}}(a; \beta a)}. \tag{8}$$

This form is the paper's key point: **data and simulation are treated symmetrically, both as Poisson counts**. The MC enters as if the template were an independent measurement of a events whose expectation is βa. Poisson-based likelihoods are known to behave better than Gaussian ones for small counts, so this likelihood is expected to outperform Conway's when the MC is small, at the same computational cost.

Weighted MC and the effective count

MC events usually carry weights. Following Bohm and Zech [5], a sum of weights $\sum_i w_i$ is treated as a scaled Poisson variable with *effective count*

$$n_{\mathrm{eff}} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2}. \tag{9}$$

The paper's improved version ([1], Sec. 5) treats the whole prediction μ_0 as one template whose MC variance is V_μ, with effective count

$$a = \mu_{\mathrm{eff}} = \frac{\mu_0^2}{V_\mu}, \qquad Q = Q_{\mathrm{P}}(n; \beta\mu_0) + Q_{\mathrm{P}}(\mu_{\mathrm{eff}}; \beta\mu_{\mathrm{eff}}), \qquad \beta = \frac{n + \mu_{\mathrm{eff}}}{\mu_0 + \mu_{\mathrm{eff}}} \tag{10}$$

([1], Eqs. 17–18, with unweighted data). This removes the "one dominant component" restriction, and it is the form implemented here and in MaCh3.

What the paper finds

The authors fit two toy problems with exact Barlow–Beeston (ROOT's `TFractionFitter`), Conway, and their new likelihood, for MC samples of 100, 1000 and 10 000 events per template, and measure the bias and the coverage (pull width) of the fitted yields. Their conclusions:

- with large MC samples all three agree, because the MC uncertainty becomes negligible;
- with small, Poisson-distributed templates the approximate likelihoods perform comparably to the exact one; Conway's produces the most outliers;
- with *weighted* templates the exact Barlow–Beeston likelihood underestimates the variance of the fitted yields, while both generalised approximate likelihoods match it;
- the approximations are much faster than exact Barlow–Beeston once there are more than about 200 bins, and the new one is about 25 % faster than Conway's for few bins.

6 How MaCh3 does it

MaCh3 (github.com/mach3-software/MaCh3) offers the test statistic as a configuration choice (`LikelihoodOptions: TestStatistic`): `Poisson`, `BarlowBeeston` (Conway), `DembinskiAbdelmotteleb`, `IceCube` (the marginal likelihood of Argüelles et al. [6]) and `Pearson`. All are evaluated per analysis bin in `SampleHandlerInterface::GetTestStatLLH(data, mc, w2)`, where `w2` is the sum of the squared weights of all MC events (signal and background) filled into the bin. By default `w2` is computed once and then frozen (`UpdateW2 = false`), while the effective count $k = \mu^2/w_2$ is recomputed from the current prediction at every call: MaCh3 holds the *absolute* MC uncertainty fixed. For a given prediction and effective count, the fitter's per-bin ("independent") option evaluates exactly MaCh3's formulas; this is checked in `tests/test_mcstat.jl`.

7 Implementation in SolarOscFitter

7.1 Effective MC counts

For each channel and reconstructed-energy bin j, the prediction at the Asimov truth, summed over day and all night bins, M_j, is rebuilt from its components, and its MC variance V_j is computed (`mc_effective_counts` in `src/likelihoods/mcstat.jl`). The effective count is $a_j = M_j^2/V_j$, computed once at the truth and then held fixed. Holding a_j fixed means holding the *relative* MC uncertainty $1/\sqrt{a_j}$ fixed as the prediction moves; MaCh3 holds the absolute one fixed instead (Sec. 6). The two agree at the truth. For a pure normalisation change (e.g. a flux scale), the relative uncertainty of a reweighted MC sample is indeed unchanged, which is why this choice was made.

Backgrounds. Each template's bin content is a sum of selected event weights, so its relative variance is $\sum w^2/(\sum w)^2$; it is multiplied by the square of the template's contribution at its true normalisation.

Signal. The response $R_{ij} = n_{ij}/N_i$ is filled with n_{ij} MC events and normalised per true-energy row ($N_i = \sum_j n_{ij}$). Given N_i a row is multinomial, so $\mathrm{Var}(n_{ij}) = n_{ij}(1 - n_{ij}/N_i)$. An MC event of row i stands for T_i/N_i predicted events, where T_i is the oscillated true-energy rate (day + night), so

$$V_j^{\mathrm{resp}} = \sum_i n_{ij}\Big(1 - \frac{n_{ij}}{N_i}\Big)\Big(\frac{T_i}{N_i}\Big)^2(\varepsilon_j g_j)^2, \tag{11}$$

with ε_j the selection efficiency and g_j a fixed per-bin factor (forward fraction with the scattering-angle cut, semi-inclusive split). The efficiency, a ratio of MC counts $\varepsilon_j = s_j/t_j$, adds the binomial term $c_j^2(1-\varepsilon_j)/(\varepsilon_j t_j)$.

Modes. Plain, angular, inclusive and semi-inclusive channels are covered; the semi-inclusive mis-identification term enters M_j without MC variance.

The components are checked to reproduce the Asimov data bin by bin, in every mode (`tests/derived_mode.jl`), and the signal variance is checked against brute-force resampling of the MC (`tests/test_mcstat.jl`).

7.2 One factor per energy bin for day and night

The night prediction in each zenith bin is built from the same MC events as the day prediction. For backgrounds it is exactly the day template weighted by exposure; for signal the same response matrix is weighted by different (oscillated) true-energy rates, so the correlation is close to, but not exactly, one. Their MC fluctuations are therefore not independent: if the MC happened to put too many events into energy bin j, it did so for the day bin and for every night bin. Giving each $(\cos z, E)$ bin its own factor, as a literal per-bin treatment does, counts some 50 copies of one fluctuation as independent. That has two effects: it *understates* the uncertainty on the overall normalisation of the energy bin, and it *overstates* the MC freedom in the day/night shape, since each zenith bin can then move on its own and absorb part of a day-night difference, which is where most of the Δm^2_{21} information is. The default (`mc_stat_day_night: shared`) therefore uses one β_j for the group G_j of all day and night bins of energy bin j:

$$Q_j = \sum_{b\in G_j} Q_{\mathrm{P}}(n_b; \beta_j\mu_b) + Q_{\mathrm{P}}(a_j; \beta_j a_j) \quad \text{(Dembinski–Abdelmotteleb)}. \tag{12}$$

Because $\partial Q_{\mathrm{P}}(n;\beta\mu)/\partial\beta = 2(\mu - n/\beta)$ is linear in $1/\beta$, the minimisation stays in closed form:

$$\sum_b\Big(\mu_b - \frac{n_b}{\beta}\Big) + a\Big(1 - \frac{1}{\beta}\Big) = 0 \;\Rightarrow\; \beta_j = \frac{N_j + a_j}{M_j + a_j}, \qquad N_j = \sum_{b\in G_j} n_b,\; M_j = \sum_{b\in G_j}\mu_b. \tag{13}$$

For Conway's form, $Q_j = \sum_b Q_{\mathrm{P}}(n_b; \beta\mu_b) + a_j(\beta-1)^2$, the same step gives the quadratic $a_j\beta^2 + (M_j - a_j)\beta - N_j = 0$. With $\beta = 1 + x$ both are evaluated in a form that does not lose precision when a_j is large:

$$\text{DA:}\quad x = \frac{N_j - M_j}{M_j + a_j}, \qquad \tfrac{1}{2}Q_{\mathrm{P}}(a; \beta a) = a_j\left(x - \ln(1+x)\right), \tag{14}$$

$$\text{Conway:}\quad x = \frac{-2(M_j - N_j)}{(a_j + M_j) + \sqrt{(a_j - M_j)^2 + 4a_jN_j}}, \quad \tfrac{1}{2}a_j(\beta-1)^2 = \tfrac{1}{2}a_jx^2. \tag{15}$$

No iterative solver is needed. The likelihood remains smooth and is differentiated exactly by ForwardDiff and by Enzyme. In Conway's form, a group with no data ($N_j = 0$) and $M_j > a_j$ has its optimum at the boundary $\beta_j = 0$ (the Gaussian constraint is weaker than the data term); the group then has zero gradient. This is a known property of Conway's approximation that the Dembinski–Abdelmotteleb form does not share, since $\beta_j = a_j/(M_j + a_j) > 0$.

7.3 Properties checked by the tests

- the closed-form β is the profile minimum (against brute-force minimisation over β), for single bins and for shared groups;
- the per-bin form equals MaCh3's `GetTestStatLLH` for both methods;
- $a \to \infty$ recovers the plain Poisson likelihood; with Asimov data $\beta = 1$ and the likelihood is unchanged at the truth;
- profiling can only raise the likelihood relative to Poisson, and less so for larger a;
- the per-bin diagnostic pieces add up to the channel likelihood;
- the full likelihood with each option is pinned by fixture regressions, and Enzyme agrees with ForwardDiff.

7.4 Usage

key	values	meaning
`mc_stat_likelihood`	`Poisson` (default)	no MC uncertainty
	`DembinskiAbdelmotteleb`	Eq. (10) (recommended)
	`BarlowBeeston`	Conway, Eq. (6)
`mc_stat_day_night`	`shared` (default)	one β per energy bin, day + night
	`independent`	one β per bin (MaCh3's per-bin treatment)

Both keys can be set at the top level of the configuration (default for every detector) or in a detector block. The setup log reports the median and maximum relative MC uncertainty $1/\sqrt{a_j}$ in the fit window, and the per-bin breakdown CSV lists a_j.

7.5 Limitations

- The effective counts (relative MC uncertainties) are frozen at the truth. Away from the truth, the MC variance of a reweighted prediction changes in ways a single relative uncertainty does not capture; MaCh3's default freezes the absolute one instead.
- Correlations of one MC sample across channels are not modelled: in semi-inclusive mode the same CC MC feeds the ES channel (forward fraction) and the CC channel (backward fraction), each with its own β; the ES MC also feeds the mis-identification term.
- One β per energy bin covers all its scattering-angle bins, whose composition (forward- peaked ES signal, flat CC and background) differs, so per-component shape fluctuations within the bin are not modelled.
- The signal efficiency's MC uncertainty is treated as independent of the response's; the semi-inclusive mis-identification response and the scattering-angle response carry no MC variance.

- Pre-binned input files (histograms instead of events) have no meaningful $\sum w^2$: each row looks like a single heavy event, which makes their MC uncertainty appear huge. The inputs must be event-level.

References

[1] H. Dembinski and A. Abdelmotteleb, *A fast and stable approximate maximum-likelihood method for template fits*, arXiv:2206.12346 [stat.ME] (2022).

[2] R. Barlow and C. Beeston, *Fitting using finite Monte Carlo samples*, Comput. Phys. Commun. **77** (1993) 219.

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[6] C. A. Argüelles, A. Schneider and T. Yuan, *A binned likelihood for stochastic models*, JHEP **06** (2019) 030, arXiv:1901.04645.